

Completion Count and Median Completed Time Analysis of SES Master's and PhD students from COVID through Post-COVID by Terms (2020–2024)

Summary of Significant Differences

Degree	Term	Significant Differences?	Details	Difference in Median Completion Time (Years)	Conclusion to question: Did median time to graduate change?
MA	Spring	No	N/A	N/A	No
MA	Summer	Yes	2020 vs 2021	+0.34	Yes
		Yes	2020 vs 2023	+0.67	Yes
		Yes	2023 vs 2024	-0.34	Yes
MA	Fall	No	N/A	N/A	No
PHD	Spring	No	N/A	N/A	No
PHD	Summer	Yes	2022 vs 2023	-5.00	Yes
PHD	Fall	No	N/A	N/A	No

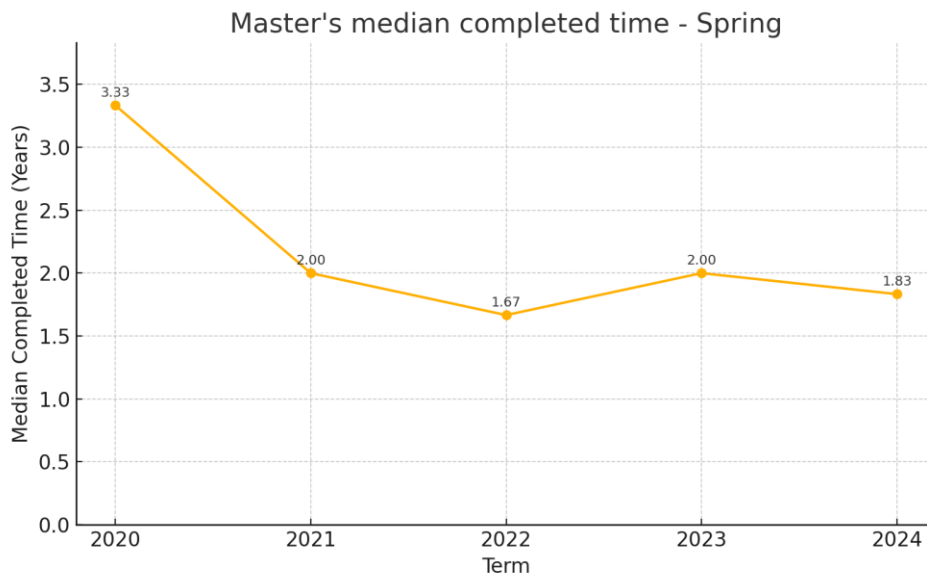
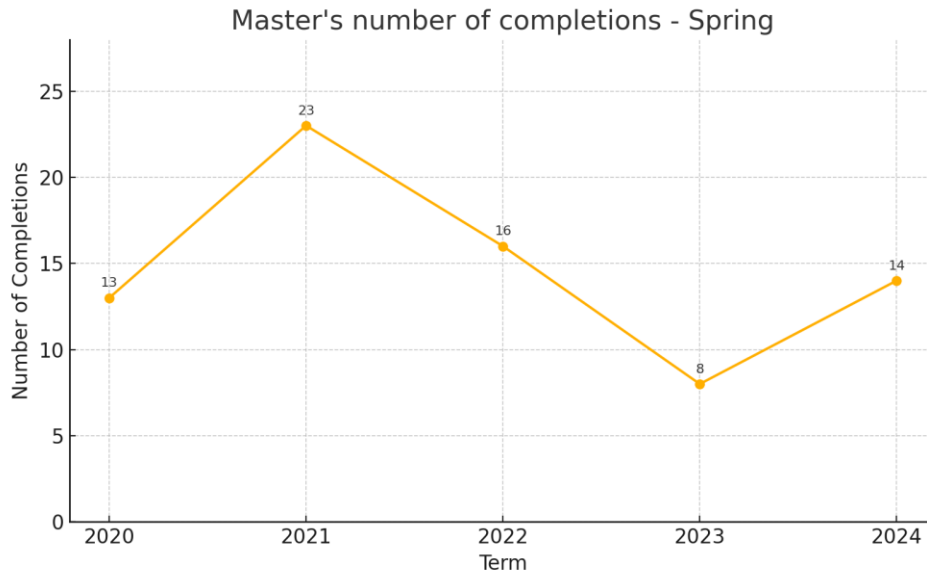
The only significant differences in Median Completion Time occurred between Summer terms (Significant Differences = “Yes”)

There was a significant difference in Median Completion Time for Master's students from 2020 to 2021 (increase of 0.34 years). Another was from 2020 to 2023 (increase of 0.67 years) and finally from 2023 to 2024 (decrease of 0.34 years).

There was a significant difference in Median Completion Time for PhD students from 2022 to 2023 (decrease of 5.00 years).

For both Master's and PhD students, differences in median completion times by term from year-to-year existed in the data for the remaining terms (Significant Differences = “No”), but were not considered statistically significant, so no general conclusion could be reached. In these cases, we assume there was no change in Median Completion Time.

Master's - Spring

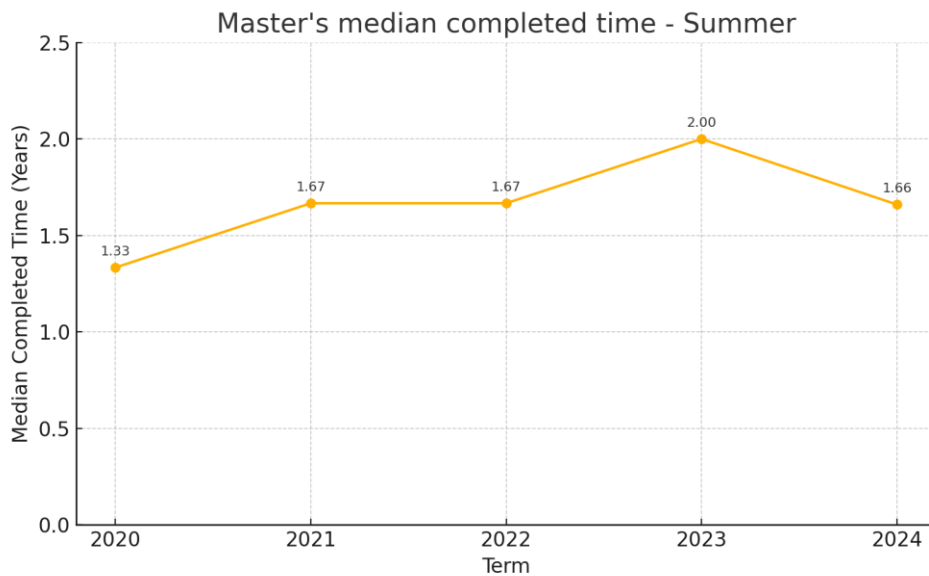
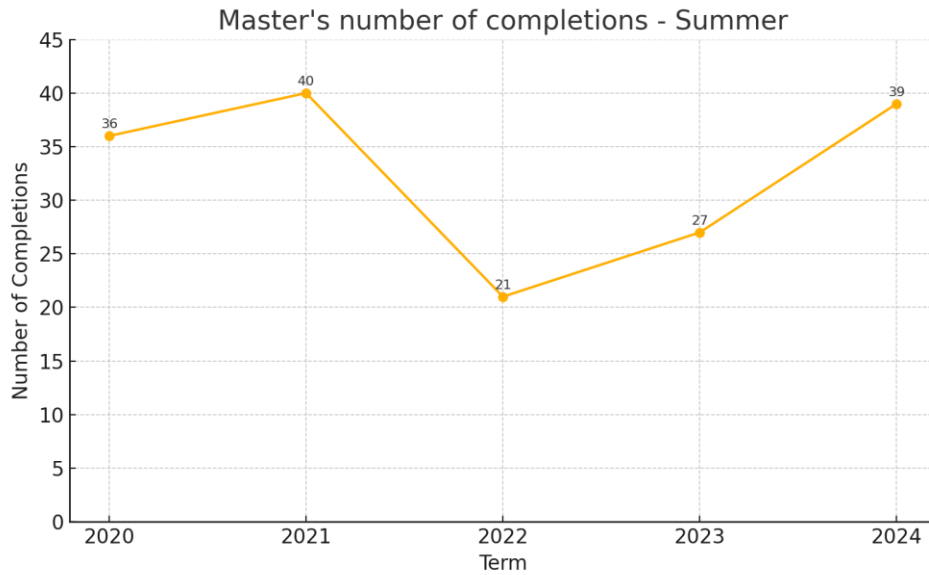


While there was an overall decrease in Median Completion Time from 2020 to 2024 for Spring Master's graduates and differences in Median Completion Time between years, there was no *statistically significant* differences found between years.

One possible reason for this uncertainty is there were not enough Completions (see Master's number of completions – Spring chart) to reliably suggest a pattern. Another possible reason is there was no actual difference.

Therefore, we cannot claim there was any actual difference in Median Completion Time over that period for Master's students graduating in the Spring.

Master's - Summer

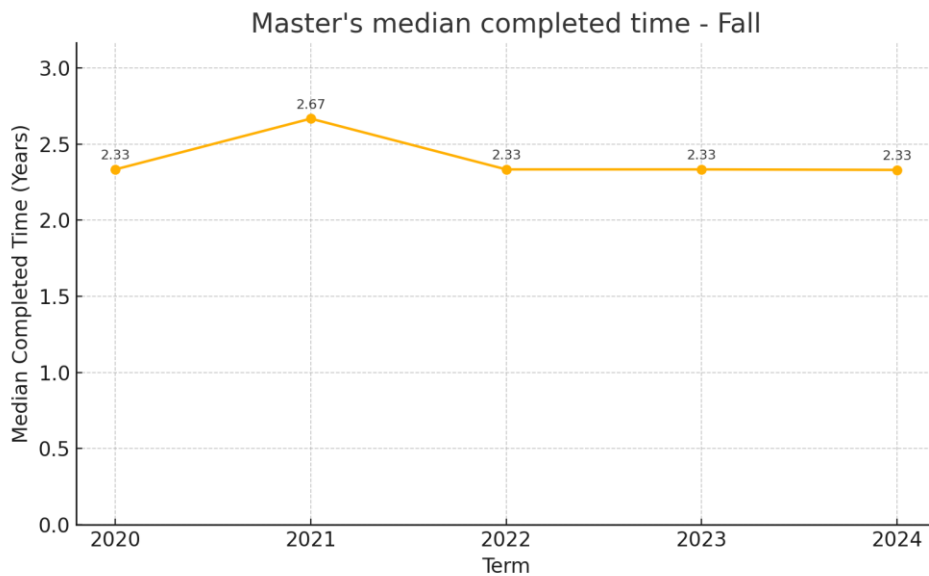
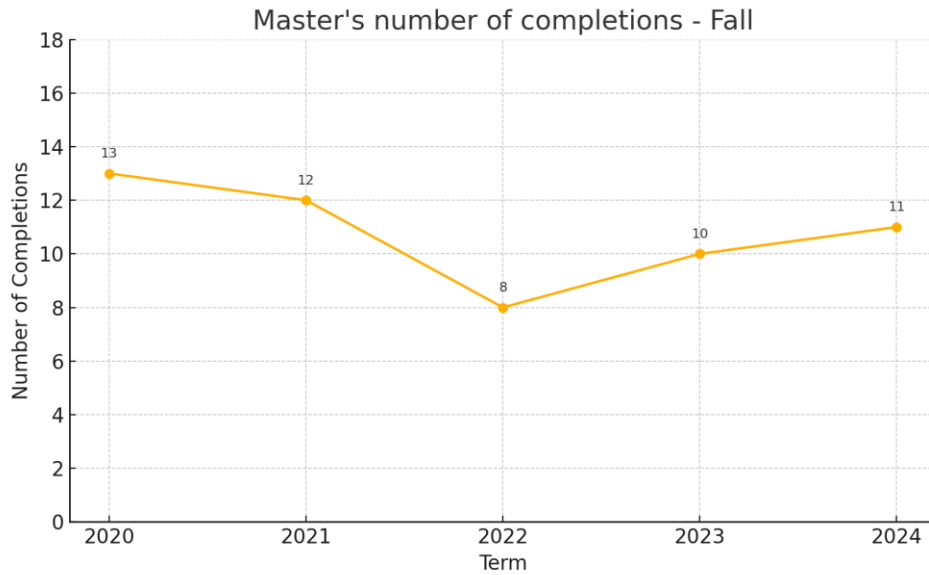


Significant differences in Median Completion Time for Summer Master's graduates were found between the following:

- - 2023 vs 2024 (-0.37 years to completion)
- - 2020 vs 2023 (+0.67 years to completion)
- - 2020 vs 2021 (+0.34 years to completion)

We can conclude that there was an increase from 2020 to 2023 in Median Completion Time, followed by a decrease from 2023 to 2024 for Master's students graduating in the Summer.

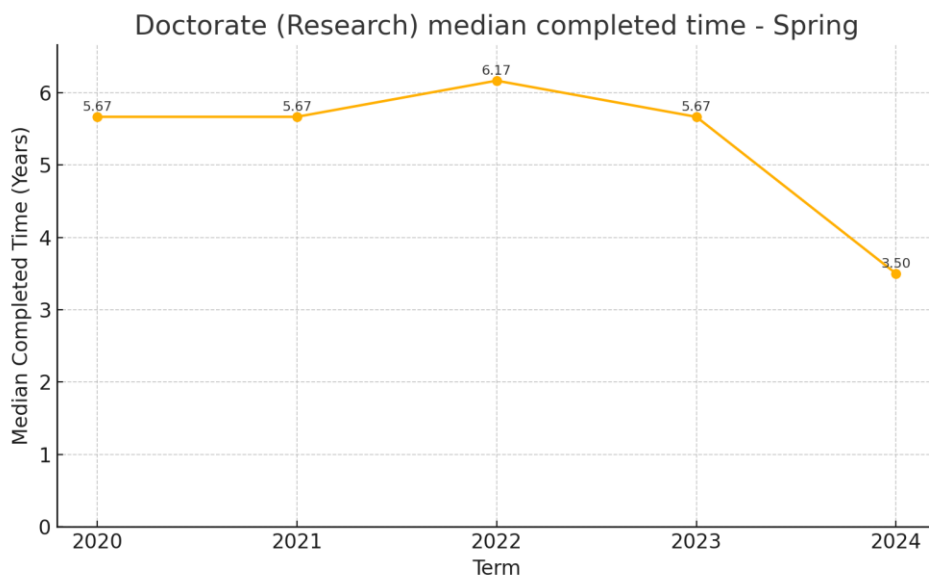
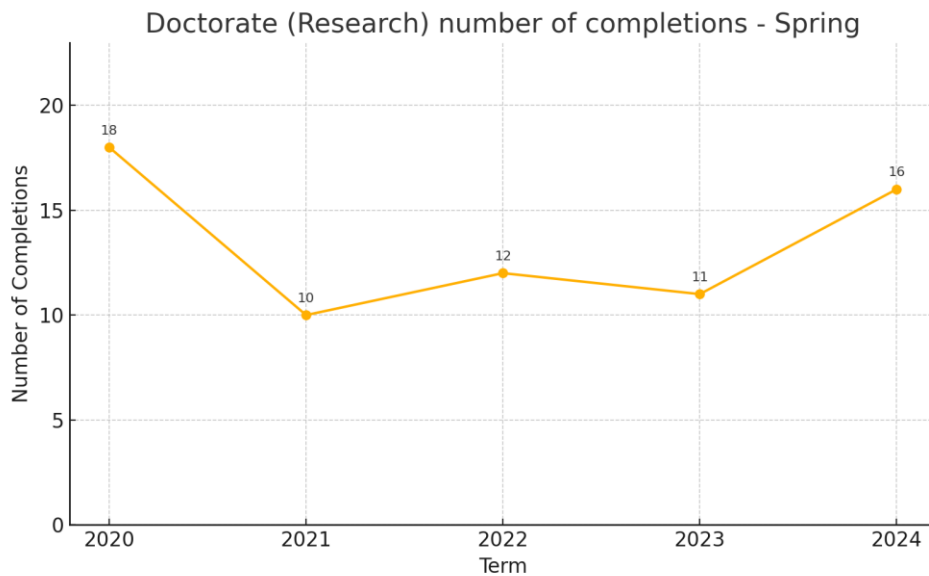
Master's - Fall



Overall, Median Completion Time from 2020 to 2024 for Fall Master's graduates remained relatively unchanged.

Therefore, we cannot claim there was any actual difference in Median Completion Time over that period for Master's students graduating in the Fall.

Doctorate (Research) - Spring

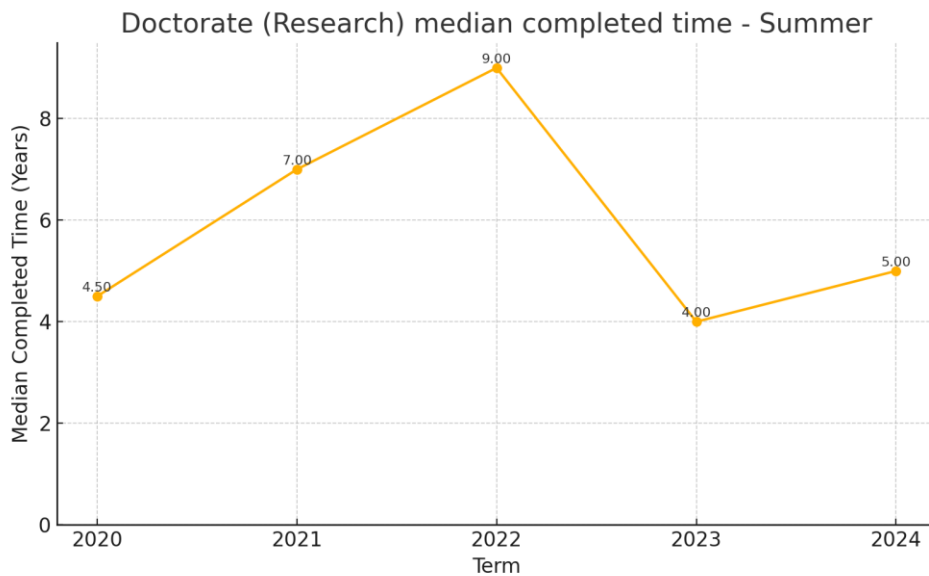
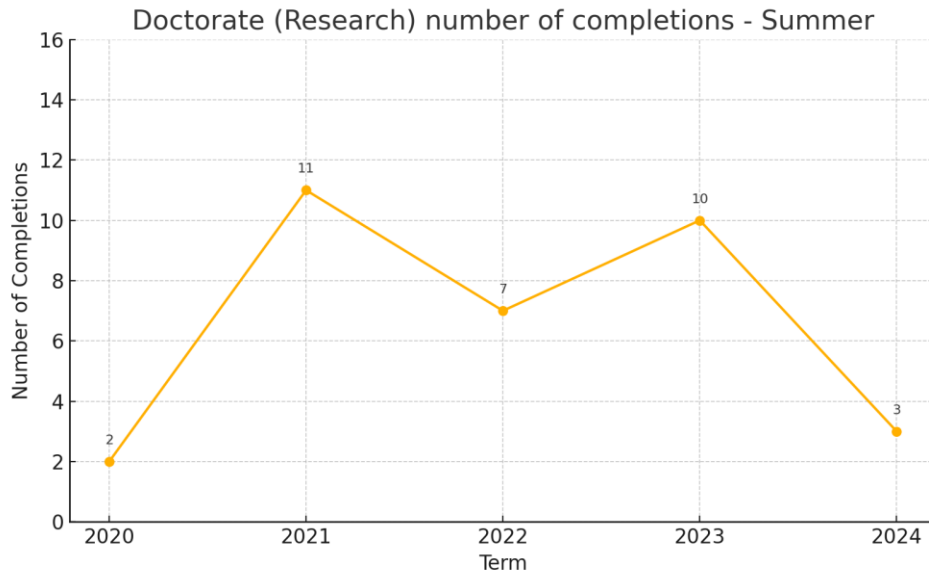


While there was an overall decrease in Median Completion Time from 2020 to 2024 for Spring PhD graduates and occasional differences in Median Completion Time between years, there wasn't a *statistically significant* differences found between years.

One possible reason for this uncertainty is there were not enough Completions (see PhD number of completions – Spring chart) to reliably suggest a pattern. Another possible reason is there was no actual difference.

Therefore, we cannot claim there was any actual difference in Median Completion Time over that period for Doctoral (Research) students graduating in the Spring.

Doctorate (Research) - Summer

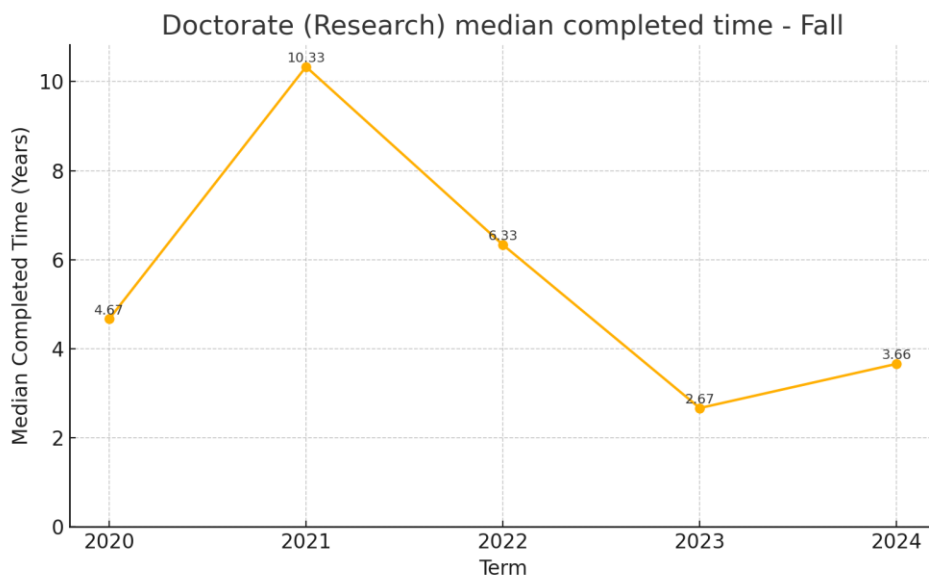
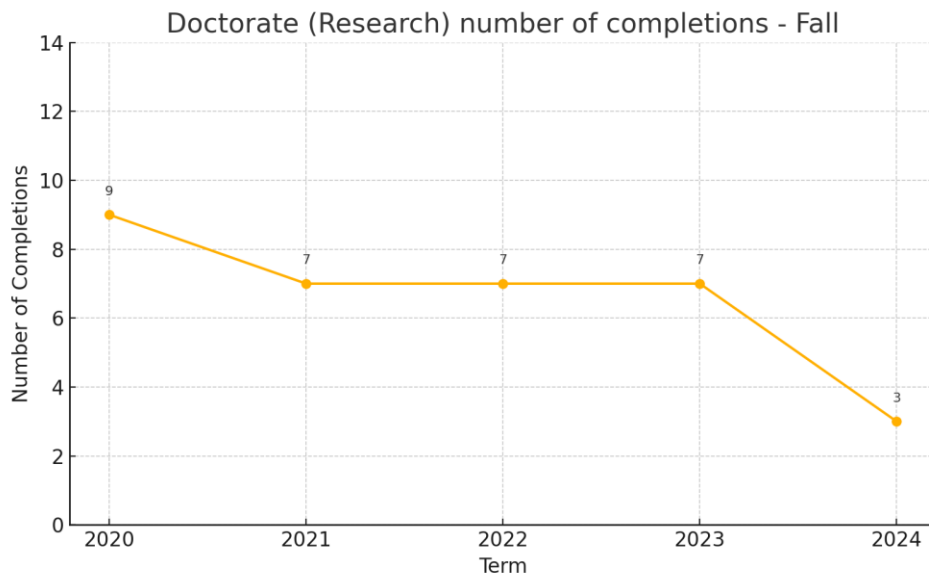


Significant differences in Median Completion Time for Summer PhD graduates were found between the following:

- - 2022 vs 2023 (-5.00 years to completion)

We can conclude that there was a decrease from 2022 to 2023 in Median Completion Time for Doctoral (Research) students graduating in the Summer.

Doctorate (Research) - Fall



While there was an overall decrease in Median Completion Time from 2020 to 2024 for Fall PhD graduates and differences in Median Completion Time between years, there was no *statistically significant* differences found between years.

One possible reason for this uncertainty is there were not enough Completions (see PhD number of completions – Fall chart) to reliably suggest a pattern. Another possible reason is there was no actual difference.

Therefore, we cannot claim there was any actual difference in Median Completion Time over that period for Doctoral (Research) students graduating in the Fall.

Appendix

		Kruskal-Wallis	
Degree	Season	Statistic	P-Value
MA	Spring	7.296864794804327	0.12100754820416339
MA	Summer	27.75091744882587	1.400990561179027 e-05
MA	Fall	1.4916603722340955	0.8281176939553737
PHD	Spring	8.971926920036406	0.061805017523580384
PHD	Summer	11.274061238794726	0.023650619927586342
PHD	Fall	9.302165	0.053974952889471454

Dunn's test on those pairs of Degrees and Seasons with $p < 0.05$

Base Year	Comparison Year	Z-Statistic	P-Value	P-Value (Bonferroni)	Degree	Season
2024	2023	-3.99707	6.413039569674872 e-05	0.000641304	MA	Summer
2023	2020	4.599167188030238	4.241832644247268 e-06	4.2418326442472676 e-05	MA	Summer
2021	2020	2.9532540876052717	0.003144	0.031444303	MA	Summer
2023	2022	-2.92148	0.003484	0.034837535199236935	PHD	Summer